

$^{32}\text{S}(\alpha, p\gamma)$ 1970Ho09, 1972Br33, 1973Al22

$J^\pi=0^+$ for ^{32}S ground state.

- 1970Ho09:** 10.975-, 11.960-, 12.120-MeV α beams were produced from the Chalk River MP tandem accelerator. Targets were 100- $\mu\text{g}/\text{cm}^2$ Sb_2S_3 on a 10- $\mu\text{g}/\text{cm}^2$ carbon backing. γ rays were detected using 15.2-cm-long by 12.7-cm-diam NaI(Tl) detectors. Protons were detected using an annular silicon surface detector close to 180° with FWHM=30 keV for protons and FWHM=50 keV for α . Measured E_p , E_γ , I_γ , $p\gamma(\theta)$. Deduced 25 level energies from E_p , J , branching ratios, multipole mixing ratios.
- 1972Br33:** a 10-MeV α beam was produced from the Chalk River MP tandem accelerator. Targets were 340- $\mu\text{g}/\text{cm}^2$ Sb_2S_3 on gold backings. γ rays were detected using a 44- cm^3 Ge(Li) detector inside an anticoincidence split annular NaI(Tl) detector. Measured E_γ , I_γ . Deduced 11 level energies, 16 γ branching ratios, 11 lifetimes using the Doppler Shift Attenuation Method (DSAM). **1970In01** from the same experiment deduced the lifetime using DSAM for a level at 3942 keV and the mixing ratio for a 3942- $\rightarrow$ 2646 γ transition.
- 1973Al22:** a 14-MeV α beam was produced from the 80-cm cyclotron at cyclotron at the Research Institute for Physics in Stockholm and 12-16-MeV α beams were produced from the Uppsala tandem accelerator. Targets were prepared by depositing pulverized sulphur enriched to 99.9% in ^{32}S on lead backings. γ rays were detected using an Ortec 60- cm^3 Ge(Li) detector with FWHM=2.2 keV at 1333 keV and an additional Canberra 60- cm^3 Ge(Li) detector for $\gamma\gamma$ -coincidence measurements. Measured E_γ , I_γ , $\gamma\gamma$ -coin, 4 $\gamma(\theta)$. Deduced 12 level energies from E_γ , J for 4 levels, branching ratios, and mixing ratios.
- 1969In04:** 6-8-MeV α beams were produced from the Chalk River MP tandem accelerator. A thick target (270 $\mu\text{g}/\text{cm}^2$) of natural sulfur on a tantalum backing and a thin target (50 $\mu\text{g}/\text{cm}^2$) of Sb_2S_3 on a gold backing were used. γ rays were detected using two Ge(Li) detectors of 15 and 40 cm^3 volume. Measured E_γ , I_γ . Deduced 6 level energies, branching ratios, 5 lifetimes using the Doppler Shift Attenuation Method (DSAM) and 1 lifetime for the 3163-keV level using the Recoil Distance Method (RDM), and transition strengths.
- 1974Lo17:** 11.0, 12.0 and 14.7-MeV α beams were produced from the Oliver Lodge Laboratory at the University of Liverpool. Targets were natural sulphur in CdS. Protons were detected using an annular ΔE -E telescope close to 180° with FWHM=80 keV. γ rays were detected using five NaI(Tl) and one Ge(Li) detectors. Measured E_γ , I_γ , $p\gamma$ -coin, $\gamma\gamma$ -coin, γ -ray linear polarization. Deduced J and π for 3 levels, 6 γ branching ratios, 5 mixing ratios, and lifetimes for a 5408-keV level using DSAM and for a 6088-keV level using RDM.
- 1973Br26:** a 11.2-MeV α beam was produced from the Legnaro (Padova) Van de Graaff generator. Targets were natural CdS. γ rays were detected using a Ge(Li) detector. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.
- 1973An01:** 5.5- and 8.6-MeV α beams of 100-200 nA were produced from the Oxford Vertical Van de Graaff (Injector) or the Oxford E(n) Tandem. Targets were natural sulphur in AgS. γ rays were detected using a Ge(Li) detector at 0° . Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

 ^{35}Cl Levels

E(level) [†]	J^π	$T_{1/2}$	Comments
0	$3/2^+$		J^π : from Adopted Levels.
1219.5 10	$1/2^+$	0.21 ps +10-8	$T_{1/2}$: from 1969In04 using DSAM. Parity from Adopted Levels.
1763.3 6	$5/2^+$	0.55 ps 15	J^π : spin=5/2 from $\gamma(\theta)$ in 1970Ho09 . Parity from Adopted Levels.
			$T_{1/2}$: from 1969In04 using DSAM.
2645.8 8	$7/2^+$	0.30 ps 9	J^π : spin=7/2 from $\gamma(\theta)$ in 1970Ho09 . Parity from 882 γ , M1+E2 to $5/2^+$ in 1970Ho09 .
			$T_{1/2}$: from 1969In04 using DSAM.
2693.6 13	$3/2, 5/2$	70 fs 30	J^π : spin=3/2, 5/2 from $\gamma(\theta)$ in 1970Ho09 .
			$T_{1/2}$: from 1972Br33 using DSAM. Other: $\tau < 30$ fs (1969In04 , DSAM).
3002.6 15	$5/2^+$	48 fs 9	J^π : spin=3/2, 5/2 from $\gamma(\theta)$ in 1970Ho09 . 1970Ho09 ruled out 3/2 by RUL. Parity from 3002.5 γ , M1+E2 to $3/2^+$ in 1970Ho09 .
			$T_{1/2}$: from 1972Br33 using DSAM. Other: $\tau < 50$ fs (1969In04 , DSAM).
3163.1 11	$7/2^-$	60 ps 7	J^π : spin=7/2 from $\gamma(\theta)$ in 1970Ho09 . Parity from 3162.6 γ , M2+E3 to $3/2^+$ in 1970Ho09 .
			$T_{1/2}$: from 1969In04 .
3915 2		<22 fs	E(level): from 1972Br33 .
			$T_{1/2}$: from 1972Br33 using DSAM.
3943.4 12		0.33 ps 5	$T_{1/2}$: from 1972Br33 using DSAM.
3968 2		<54 fs	E(level): from 1972Br33 .
			$T_{1/2}$: from 1972Br33 using DSAM.

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π	T _{1/2}	Comments
4058 3		31 fs 13	E(level): weighted average of 4045 10 (1970Ho09) and 4058 2 (1972Br33). T _{1/2} : from 1972Br33 using DSAM.
4108 2	3/2, 7/2	70 fs 16	E(level): weighted average of 4110 10 (1970Ho09) and 4108 2 (1972Br33). J ^π : from γ(θ) in 1970Ho09. T _{1/2} : from 1972Br33 using DSAM.
4171 2		68 fs 24	E(level): weighted average of 4175 10 (1970Ho09) and 4171 2 (1972Br33). T _{1/2} : from 1972Br33 using DSAM.
4347.8 12	9/2 ⁻	2.9 ps +14-7	J ^π : spin from γ(θ) in 1974Lo17. Parity from 1702γ, E1 to 7/2 ⁺ and 1184.6γ, M1+E2 to 7/2 ⁻ in 1974Lo17. Other: J=5/2, 9/2 from γ(θ) in 1970Ho09 and 1973Al22. T _{1/2} : from 1972Br33 using DSAM.
4618 2	3/2, 5/2	58 fs 24	E(level): from 1972Br33. J ^π : from γ(θ) in 1970Ho09. T _{1/2} : from 1972Br33 using DSAM.
4770 15			E(level): from 1970Ho09.
4880.8 21		0.28 ps 6	E(level): other: 4882 2 from 1972Br33, 4885 20 from 1970Ho09. T _{1/2} : from 1972Br33 using DSAM.
5015 20			E(level): from 1970Ho09.
5175 20			E(level): from 1970Ho09.
5230 20	1/2, 3/2, 5/2		E(level): from 1970Ho09. J ^π : from γ(θ) in 1970Ho09.
5407.1 15	11/2 ⁻	0.4 ps 1	E(level): other: 5415 20 from 1970Ho09. J ^π : spin from γ(θ) in 1974Lo17 and from γ(θ) and yield function in 1973Al22. Parity from 1059.3γ, M1+E2 to 9/2 ⁻ in 1974Lo17. T _{1/2} : from 1974Lo17 using DSAM.
5535 20			E(level): from 1970Ho09.
5600 20			E(level): from 1970Ho09.
5650 20			E(level): from 1970Ho09.
5850 25	5/2, 9/2		E(level): from 1970Ho09. J ^π : from γ(θ) in 1970Ho09.
5927.1 19	7/2, 11/2		E(level): other: 5930 25 from 1970Ho09. J ^π : from γ(θ) in 1973Al22.
6087.0 18	13/2 ⁻	9.3 ps 8	J ^π : spin from γ(θ) in 1974Lo17. Other: J=9/2, 13/2 from γ(θ) in 1973Al22. Parity from 679.9γ, M1(+E2) to 11/2 ⁻ in 1974Lo17. T _{1/2} : from 1974Lo17 using RDM.
6400 25			E(level): from 1970Ho09.

[†] From a least-squares fit to E_γ values with uncertainties, unless otherwise noted. 1970Ho09 reported E(level) with uncertainties derived from measured E_p.

 $\gamma(^{35}\text{Cl})$

E _i (level)	J _i ^π	E _γ [†]	I _γ	E _f	J _f ^π	Mult.	δ	Comments
1219.5	1/2 ⁺	1219.5 10	100	0	3/2 ⁺			E _γ : weighted average of 1219.3 6 (1969In04) and 1219.5 10 (1973Al22).
1763.3	5/2 ⁺	543.8 1763.2 6	<0.5 100	1219.5 0	1/2 ⁺ 3/2 ⁺	M1+E2	+3.0 1	I _γ : from 1969In04. E _γ : from 1969In04. Other: 1763.1 10 (1973Al22). I _γ : from 1969In04, 1970Ho09, 1973Al22. Mult., δ: D+Q from γ(θ) in 1970Ho09. M2 ruled out by RUL.
2645.8	7/2 ⁺	882.4 10	11.1 22	1763.3	5/2 ⁺	M1+E2	+0.25 5	E _γ : from 1973Al22. I _γ : from 1970Ho09. Other: 18 (1969In04), 10 (1973Al22).

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	Mult.	δ	Comments
2645.8	7/2 ⁺	1426.3 2646.2 15	<1.2 100.0 22	1219.5 0	1/2 ⁺ 3/2 ⁺			Mult., δ : D+Q from $\gamma(\theta)$ in 1970Ho09. M2 ruled out by RUL. I $_\gamma$: from 1969In04. E $_\gamma$: from 1969In04. Other: 2646.0 20 (1973Al22). I $_\gamma$: from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
2693.6	3/2, 5/2	930.5 15 1474.1 2693.2 20	18.2 26 11.7 26 100 4	1763.3 1219.5 0	5/2 ⁺ 1/2 ⁺ 3/2 ⁺	D+Q		E $_\gamma$: from 1973Al22. I $_\gamma$: from 1970Ho09. I $_\gamma$: from 1970Ho09. Other: 10 (1969In04). E $_\gamma$: from 1973Al22. I $_\gamma$: from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ : +0.17 8 for J=3/2; -0.26 5 for J=5/2 (1970Ho09).
3002.6	5/2 ⁺	309.0 356.8 1239.3 1783.1 3002.5 15	<5 <1 <2 <2 100	2693.6 2645.8 1763.3 1219.5 0	3/2, 5/2 7/2 ⁺ 5/2 ⁺ 1/2 ⁺ 3/2 ⁺	M1+E2	+0.09 3	I $_\gamma$: from 1969In04. I $_\gamma$: from 1969In04. E $_\gamma$: from 1972Br33. I $_\gamma$: from 1969In04. Other: <4 (1972Br33). I $_\gamma$: from 1969In04, 1972Br33. E $_\gamma$: weighted average of 3003 2 (1969In04), 3003 2 (1972Br33), and 3002.0 15 (1973Al22). I $_\gamma$: from 1969In04, 1970Ho09, 1972Br33, 1973Al22.
3163.1	7/2 ⁻	160.5 469.5 517.4 15 1399.8 3162.6 15	<0.54 <0.54 11.1 33 <1.1 100.0 33	3002.6 2693.6 2645.8 1763.3 0	5/2 ⁺ 3/2, 5/2 7/2 ⁺ 5/2 ⁺ 3/2 ⁺	M2+E3	-0.14 2	Mult., δ : D+Q from $\gamma(\theta)$ in 1970Ho09. M2 ruled out by RUL. I $_\gamma$: from 1969In04. I $_\gamma$: from 1969In04. E $_\gamma$: from 1973Al22. I $_\gamma$: from 1970Ho09. Other: 8.7 (1969In04), 11 (1973Al22). I $_\gamma$: from 1969In04. E $_\gamma$: from 1973Al22. I $_\gamma$: from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
3915		2695 3915	19 5 100 5	1219.5 0	1/2 ⁺ 3/2 ⁺			Mult., δ : Q+O from $\gamma(\theta)$ in 1970Ho09. M3 ruled out by RUL. I $_\gamma$: from 1972Br33.
3943.4		1297.4 15 2180.2 15	8.1 21 100.0 21	2645.8 1763.3	7/2 ⁺ 5/2 ⁺			I $_\gamma$: from 1972Br33. E $_\gamma$: from 1973Al22. I $_\gamma$: weighted average of 9.9 22 (1970Ho09) and 6.4 21 (1972Br33). Other: 11 (1973Al22). E $_\gamma$: from 1973Al22. I $_\gamma$: from 1972Br33. Other: 100.0 22 (1970Ho09), 100 (1973Al22).
3968		2748 3968	100 <2	1219.5 0	1/2 ⁺ 3/2 ⁺			I $_\gamma$: from 1972Br33.
4058		2295 2838 4058	8 4 100 4 <2.2	1763.3 1219.5 0	5/2 ⁺ 1/2 ⁺ 3/2 ⁺			I $_\gamma$: from 1972Br33. I $_\gamma$: from 1972Br33. I $_\gamma$: from 1972Br33.
4108	3/2, 7/2	2345 4108	16 5 100 5	1763.3 0	5/2 ⁺ 3/2 ⁺			I $_\gamma$: from 1972Br33.
4347.8	9/2 ⁻	1184.6 10	100.0 31	3163.1	7/2 ⁻	M1+E2	-0.36 3	E $_\gamma$: from 1973Al22. I $_\gamma$: from 1974Lo17. Others: 100 7

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	I_γ	E_f	J_f^π	Mult.	δ	Comments
4347.8	9/2 ⁻	1702.1 15	52 7	2645.8	7/2 ⁺	E1	-0.00 3	(1970Ho09) and 100 4 (1972Br33). Mult.: D+Q from $\gamma(\theta)$ in 1970Ho09 and 1973Al22. E1+M2 ruled out by RUL. δ : from 1974Lo17. Other: -0.40 +8-9 (1973Al22); +0.42 4 (1970Ho09). $A_2=-0.95$ 3, $A_4=0.08$ 3 (1974Lo17). POL=-0.005 40 (1974Lo17). E_γ : from 1973Al22. I_γ : unweighted average of 61 7 (1970Ho09), 39 4 (1972Br33), and 56.3 31 (1974Lo17). Mult.: from $\gamma(\theta)$ and $\gamma(\text{pol})$ in 1974Lo17. δ : from 1974Lo17. $A_2=-0.34$ 5, $A_4=0.04$ 5 (1974Lo17). POL=0.36 7 (1974Lo17). E_γ, I_γ : from 1972Br33. δ : +0.08 14 for J=3/2; -0.36 15 for J=5/2 (1970Ho09).
4618	3/2, 5/2	4618 2	100	0	3/2 ⁺			Final level is uncertain, either 2646 or 2694 (1970Ho09).
4880.8		2187.1 $\ddagger$	52 $\ddagger$ 9	2693.6	3/2, 5/2			Final level is uncertain, either 2646 or 2694 (1970Ho09).
		2234.9 $\ddagger$	52 $\ddagger$ 9	2645.8	7/2 ⁺			I_γ : from 1970Ho09. E_γ : from 1973Al22. I_γ : from 1970Ho09. Other: 100 (1973Al22).
		3117.4 20	100 9	1763.3	5/2 ⁺			I_γ : from 1970Ho09. E_γ : from 1973Al22.
5230	1/2, 3/2, 5/2	5230	100	0	3/2 ⁺			I_γ : from 1970Ho09.
5407.1	11/2 ⁻	1059.3 10	9.9 22	4347.8	9/2 ⁻	M1+E2	+0.25 8	E_γ : from 1973Al22. I_γ, δ : from 1974Lo17. Mult.: D+Q from $\gamma(\theta)$ in 1970Ho09 and 1973Al22. M2 ruled out by RUL. $A_2=0.19$ 11, $A_4=-0.01$ 13 (1974Lo17). POL=-0.40 14 (1974Lo17). E_γ : from 1973Al22. I_γ, δ : from 1974Lo17. Other δ : +0.02 9 (1973Al22). Mult.: Q from $\gamma(\theta)$ in 1973Al22 and 1974Lo17. M2 ruled out by RUL. $A_2=0.44$ 3, $A_4=-0.19$ 3 (1974Lo17). δ : -2.2 5 or -0.6 1 for J=5/2; +0.28 4 for J=9/2 (1970Ho09).
		2244 3	100.0 22	3163.1	7/2 ⁻	E2	0.00 3	E_γ : from 1973Al22. I_γ, δ : from 1974Lo17. Other δ : +0.02 9 (1973Al22). Mult.: Q from $\gamma(\theta)$ in 1973Al22 and 1974Lo17. M2 ruled out by RUL. $A_2=0.44$ 3, $A_4=-0.19$ 3 (1974Lo17). δ : -2.2 5 or -0.6 1 for J=5/2; +0.28 4 for J=9/2 (1970Ho09).
5850	5/2, 9/2	2687	100	3163.1	7/2 ⁻			E_γ : from 1973Al22.
5927.1	7/2, 11/2	1579.3 15	100	4347.8	9/2 ⁻			δ : +1.1 5 for J=7/2; -0.8 4 for J=11/2 (1973Al22). E_γ : from 1973Al22.
6087.0	13/2 ⁻	679.9 10	100 6	5407.1	11/2 ⁻	M1(+E2)	+0.020 15	I_γ, δ : from 1974Lo17. Other δ : 0.00 7 for J _i =13/2 (1973Al22). Mult.: D(+Q) from $\gamma(\theta)$ in 1974Lo17. M from $\gamma(\text{pol})$ in 1974Lo17.

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$^{32}\text{S}(\alpha, \text{p}\gamma)$ [1970Ho09](#), [1972Br33](#), [1973Al22](#) (continued) $\gamma(^{35}\text{Cl})$ (continued)

<u>$E_i(\text{level})$</u>	<u>E_γ</u> [†]	<u>I_γ</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
					$A_2 = -0.22$ 1, $A_4 = 0.005$ 15 (1974Lo17). POL = -0.45 5 (1974Lo17). I_γ : from 1974Lo17 .
6087.0	1739.2	18 6	4347.8	9/2 ⁻	

[†] [1973Al22](#) is the only literature in this dataset that reported measured E_γ with uncertainties. [1969In04](#) and [1972Br33](#) reported $E(\text{level})$ with uncertainties derived from their measured E_γ but did not report E_γ . Evaluators adopt the uncertainties of $E(\text{level})$ from [1969In04](#) and [1972Br33](#) as the uncertainties of an E_γ if this γ transition populates the ground state and is also the dominant decay branch of that level. Other E_γ values without uncertainties are from level-energy differences.

[‡] Multiply placed with undivided intensity.

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Level Scheme

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given

